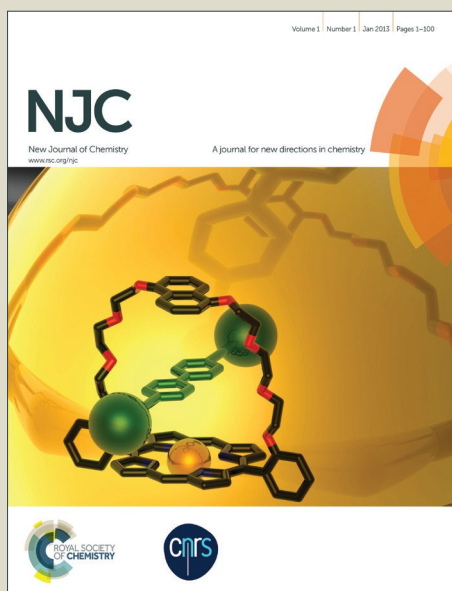


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Synthesis of Enantiomerically Pure Bis(2,2-dimethyl-1,3-dioxolanylmethyl)chalcogenides and Dichalcogenides

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Herein we describe the reaction between nucleophilic species of chalcogenium (sulfur, selenium and tellurium) and (*R*)- and (*S*)-solketal tosylate to prepare enantiomerically pure bis-1,3-dioxolanylmethyl chalcogenides and dichalcogenides in short reaction times and good yields. Furthermore, the new diselenide **4a** was evaluated in the Michael reaction with acrylonitrile, giving (*R*)-3-[(2,3-dihydroxypropyl)selenyl]propanenitrile **6b** after two steps.

1 Introduction

Organochalcogen compounds have been extensively studied due to their pharmacological and biological activities,¹ applied in materials² and as versatile building blocks for organic synthesis.³ Concerning to diaryl chalcogenides and dichalcogenides (sulfides, selenides and tellurides), several methods have been recently described towards their synthesis.⁴ However, only few studies were reported for the synthesis of chiral organochalcogenides.⁵

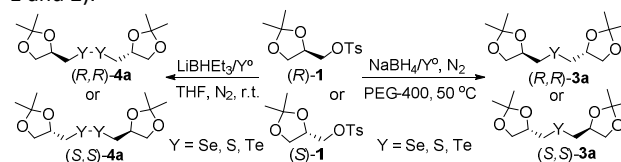
The application of chiral chalcogenides and dichalcogenides have grown over the past few years due to the versatility of these compounds in asymmetric synthesis.⁶ For example, they can be used as catalysts in the enantioselective addition of diethyl zinc to aldehydes^{6h} and in asymmetric Baylis-Hillman reactions.^{6a} In particular, water-soluble selenosugars have exhibited interesting biological activities, such as antioxidant and glycosidase inhibitors,⁷ and as potent water-soluble scavengers of HOX (X= Cl, Br, SCN).⁸

On the other hand, among the many uses of solketal,⁹ the intermediate tosyl solketal stands out due to its use as building block in organic synthesis, including the preparation of bioactive lipopeptides,^{9c} spiroheterocycles,^{9d} in the synthesis of new indole-based MK2 inhibitors¹⁰ and (*R*)- and (*S*)-epichlorohydrin.^{9e} More recently, tosyl solketal was utilized as the starting material for the preparation of novel 1,2,4-triazoles brassinosteroid biosynthesis inhibitors,^{9f} and of a new maleimide, potent inhibitor of protein GSK-3β, a key kinase target in treating osteoporosis.¹¹

In a previous paper, we described a new method for the synthesis of thioethers starting from tosyl solketal using KF/Al₂O₃ and PEG-400 as the solvent.¹² The reactions were performed under mild conditions with a range of aliphatic and aromatic thiols and it was possible to reuse the catalytic system without previous treatment and with comparable activity for one additional reaction. More recently, we describe a simple method for the synthesis of glycerol derivatives

containing an organochalcogen unit (sulfur, selenium and tellurium) using NaBH₄ and PEG-400 as the solvent.¹³ Among the compounds synthesized, 2,2-dimethyl-4-(phenyltellanylmethyl)-1,3-dioxolane proved to be a promising antioxidant.

Based on our previous works and following our research trajectory in the direction of new semi-synthetic organochalcogen compounds, we describe here the synthesis of enantiomerically pure bis-(2,2-dimethyl-1,3-dioxolanylmethyl)chalcogenides and dichalcogenides (Schemes 1 and 2).



Scheme 1. General scheme to prepare chiral chalcogenides and dichalcogenides.

2 Experimental

2.1 General Information

The reactions were monitored by TLC carried out on Merck silica gel (60 F₂₅₄) by using UV light as visualizing agent and 5% vanillin in 10% H₂SO₄ and heat as developing agents. Baker silica gel (particle size 0.040-0.063 mm) was used for flash chromatography. Proton nuclear magnetic resonance spectra (¹H NMR) were obtained at 300 MHz on a Varian Gemini NMR and at 400 MHz on Bruker DPX 400 spectrometers. Spectra were recorded in CDCl₃ and DMSO-*d*₆ solutions. Chemical shifts are reported in ppm, referenced to tetramethylsilane (TMS) as the external reference. Coupling constants (*J*) are reported in Hertz. Carbon-13 nuclear magnetic resonance spectra (¹³C NMR) were obtained at 75 MHz on a Varian Gemini NMR and at 100 MHz on Bruker DPX 400 spectrometers. Chemical shifts are reported in ppm, referenced to the solvent peak of CDCl₃ and DMSO-*d*₆. Low-resolution mass spectra were obtained with a Shimadzu GC-MS-QP2010 mass spectrometer. Mass spectra were obtained for all compounds on a LTQ Orbitrap Discovery mass spectrometer (Thermo Fisher Scientific). This hybrid system meets the LTQ XL linear ion trap mass spectrometer and an Orbitrap mass analyzer. The experiments were performed via direct infusion of sample (flow: 10 μL/min) in the positive-ion mode using electrospray ionization. Elemental composition calculations for comparison were executed using the specific tool included in the Qual Browser

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†Electronic Supplementary Information (ESI) available: Experimental details, NMR spectra and physicochemical data of all the synthesized compounds are available. See DOI: 10.1039/x0xx00000x

ARTICLE

Journal Name

module of Xcalibur (Thermo Fisher Scientific, release 2.0.7) software. Elemental analyses CHN were performed on a model 2400 Series II - Perkin Elmer elemental analysis instrument. Optical rotations were measured with a JASCO P-2000 Polarimeter in CH₂Cl₂ solutions as the solvents with percent concentrations. All enantiomeric excesses were obtained using Shimadzu GC-2010 chromatograph and Agilent 6820 GC System using the following column: HYDRODEX®-β-3P (25 m x 0.25 mm ID) - Macherey-Nagel.

2.2 Synthesis of Compounds 3, 4 and 6

2.2.1 General procedure for the direct synthesis of bis-1,3-dioxolanymethyl chalcogenides 3: To a mixture of respective chalcogen **2** (0.5 mmol) in PEG-400 (4.0 mL) under N₂ atmosphere, was added NaBH₄ (0.042g; 1.1 mmol) and the mixture was heated slowly to 50 °C and stirred for 40 min. The respective (*R*)- or (*S*)-solketal tosylate **1** (0.286g; 1.0 mmol) was then added at 50 °C. The reaction progress was followed by TLC. After the time indicated in Table 2, the solution was cooled to room temperature and water (10.0 mL) and ethyl acetate (15.0 mL) were added. The organic phase was washed with water (2x 10 mL), separated, dried over MgSO₄, and the solvent was evaporated under reduced pressure. The product was isolated by column chromatography using hexane/ethyl acetate (97:3) as eluent. All the synthesized compounds were characterized and the spectral data are listed below:

(*R,R*)-bis(2,2-Dimethyl-1,3-dioxolanymethyl)selenide (3a): Yield: 0.130 g (84%, 99.9 ee); yellowish oil; $[\alpha]_D^{20}$: + 47.47 (c 0.92, CH₂Cl₂). ¹H NMR (CDCl₃, 400 MHz): δ 4.28-4.34 (m, 2H); 4.11 (dd, *J* = 8.3 and 6.0 Hz, 2H); 3.69 (dd, *J* = 8.3 and 6.4 Hz, 2H); 2.87 (dd, *J* = 12.5 and 5.6 Hz, 2H); 2.70 (dd, *J* = 12.5 and 7.2 Hz, 2H); 1.42 (d, *J* = 0.6 Hz, 6H); 1.35 (d, *J* = 0.6 Hz, 6H). ¹³C NMR (100 MHz, CDCl₃); δ (ppm): 109.5, 75.9, 69.3, 27.1, 26.9, 25.5. MS: *m/z* (rel. int.) 310 (8.8), 101 (73.6), 43 (100.0). HRMS (ESI): *m/z* calcd for C₁₂H₂₂O₄Se [M+Na]⁺: 333.0581; found: 333.0593.

(*R,R*)-bis(2,2-Dimethyl-1,3-dioxolanymethyl)sulfide (3b): Yield: 0.098 g (75%, 99.9 ee); colorless oil; $[\alpha]_D^{20}$: + 35.13 (c 0.44, CH₂Cl₂). ¹H NMR (CDCl₃, 300 MHz): δ 4.23-4.31 (m, 2H); 4.10 (dd, *J* = 8.3 and 6.0 Hz, 2H); 3.72 (dd, *J* = 8.3 and 6.3 Hz, 2H); 2.84 (dd, *J* = 13.4 and 5.7 Hz, 2H); 2.67 (dd, *J* = 13.4 and 6.9 Hz, 2H); 1.43 (s, 6H); 1.36 (s, 6H). ¹³C NMR (100 MHz, CDCl₃); δ (ppm): 109.5, 75.5, 68.7, 35.7, 26.8, 25.4. MS: *m/z* (rel. int.) 262 (3.7), 101 (100.0), 43 (81.4). HRMS (ESI): *m/z* calcd for C₁₂H₂₂O₄S [M+H]⁺: 263.1317; found: 263.1312.

(*R,R*)-bis(2,2-Dimethyl-1,3-dioxolanymethyl)telluride (3c): Yield: 0.108 g (60%, 99.9 ee); yellowish oil; $[\alpha]_D^{20}$: + 28.95 (c 0.70, CH₂Cl₂). ¹H NMR (CDCl₃, 400 MHz): δ 4.30-4.36 (m, 2H), 4.14 (dd, *J* = 8.2 and 5.9 Hz, 2H), 3.62 (dd, *J* = 8.2 and 6.7 Hz, 2H), 2.95 (dd, *J* = 12.1 and 5.6 Hz, 2H), 2.78 (dd, *J* = 12.1 and 7.3 Hz, 2H), 1.43 (d, *J* = 0.6 Hz, 6H), 1.34 (d, *J* = 0.7 Hz, 6H). ¹³C NMR (100 MHz, CDCl₃); δ (ppm): 109.7, 76.9, 70.5, 27.0, 25.7, 6.6. MS: *m/z* (rel. int.) 360 (18.3), 101 (23.3), 43 (100.0). HRMS (ESI): *m/z* calcd for C₁₂H₂₂O₄Te [M+H]⁺: 361.0659; found: 361.0653.

(*S,S*)-bis(2,2-Dimethyl-1,3-dioxolanymethyl)selenide (3a): Yield: 0.109 g (70%, 99.9 ee); yellowish oil; $[\alpha]_D^{20}$: - 48.60 (c

1.08, CH₂Cl₂). The characterization data from NMR, MS and HRMS spectra were identical in all aspects with those of (*R,R*)-(+)-**3a** enantiomer.

(*S,S*)-bis(2,2-Dimethyl-1,3-dioxolanymethyl)sulfide (3b): Yield: 0.085 g (65%, 99.9 ee); colorless oil; $[\alpha]_D^{20}$: - 48.58 (c 0.45, CH₂Cl₂). The characterization data from NMR, MS and HRMS spectra were identical in all aspects with those of (*R,R*)-(+)-**3b** enantiomer.

(*S,S*)-bis(2,2-Dimethyl-1,3-dioxolanymethyl)telluride (3c): Yield: 0.099 g (55%, 99.9 ee); yellowish oil; $[\alpha]_D^{20}$: - 32.25 (c 0.56, CH₂Cl₂). The characterization data from NMR, MS and HRMS spectra were identical in all aspects with those of (*R,R*)-(+)-**3c** enantiomer.

2.2.2 General procedure for the direct synthesis of bis-1,3-dioxolanymethyl dichalcogenides 4: In a two-necked round-bottomed flask containing a suspension of the elemental chalcogenium **2** (2.0 mmol) in THF (2.0 mL) under N₂ atmosphere, lithium triethyl borohydride (2.0 mmol; 2.0 mL of a sol. 1M in THF) was added. The reaction mixture was stirred for 30 min at room temperature. The respective (*R*)- or (*S*)-solketal tosylate **1** (0.228g; 0.85 mmol) was then added as a solution in THF (7.5 mL) at the reaction mixture. The reaction progress was followed by TLC. After the complete consumption of starting material, saturated aqueous NH₄Cl (25.0 mL) was added to the reaction mixture. Then, the solution was diluted with dichloromethane (30 mL) and washed with water (3x 10 mL). The organic phase was separated, dried over MgSO₄, and the solvent was evaporated under reduced pressure. The product was isolated by column chromatography using hexanes/ethyl acetate (97:3) as eluent. All the synthesized compounds were characterized and the spectral data are listed below:

(*R,R*)-bis(2,2-Dimethyl-1,3-dioxolanymethyl)diselenide (4a): Yield: 0.133 g (80%, 99.9 ee); orangish oil; $[\alpha]_D^{20}$: + 34.01 (c 0.48, CH₂Cl₂). ¹H NMR (CDCl₃, 400 MHz): δ 4.34-4.41 (m, 2H); 4.14 (dd, *J* = 8.4 and 6.0 Hz, 2H); 3.74 (dd, *J* = 8.4 and 6.1 Hz, 2H); 3.20 (dd, *J* = 12.4 and 5.8 Hz, 2H); 3.07 (dd, *J* = 12.4 and 7.1 Hz, 2H); 1.44 (s, 6H); 1.36 (s, 6H). ¹³C NMR (100 MHz, CDCl₃); δ (ppm): 109.6, 75.9, 69.0, 33.0, 27.0, 25.5. MS: *m/z* (rel. int.) 390 (10.5), 115 (76.5), 101 (21.2), 57 (97.5), 43 (100.0). HRMS (ESI): *m/z* calcd for C₁₂H₂₂O₄Se₂ [M+H]⁺: 390.9927; found: 390.9921.

(*R,R*)-bis(2,2-Dimethyl-1,3-dioxolanymethyl)disulfide (4b): Yield: 0.052 g (42%, 99.9 ee); colorless oil; $[\alpha]_D^{20}$: + 24.66 (c 0.52, CH₂Cl₂). ¹H NMR (CDCl₃, 300 MHz): δ 4.33-4.41 (m, 2H); 4.14 (dd, *J* = 8.4 and 6.0 Hz, 2H); 3.76 (dd, *J* = 8.4 and 6.0 Hz, 2H); 2.98 (dd, *J* = 13.5 and 5.7 Hz, 2H); 2.82 (dd, *J* = 13.5 and 7.0 Hz, 2H); 1.43 (s, 6H); 1.36 (s, 6H). ¹³C NMR (75 MHz, CDCl₃); δ (ppm): 109.6, 74.7, 68.6, 42.1, 26.9, 25.5. MS: *m/z* (rel. int.) 294 (11.8), 101 (91.7), 43 (100.0). HRMS (ESI): *m/z* calcd for C₁₂H₂₂O₄S₂ [M+H]⁺: 295.1032; found: 294.9932.

(*R,R*)-bis(2,2-Dimethyl-1,3-dioxolanymethyl)ditelluride (4c): Yield: 0.098 g (47%, 99.9 ee); reddish oil; $[\alpha]_D^{20}$: + 30.13 (c 0.54, CH₂Cl₂). ¹H NMR (CDCl₃, 400 MHz): δ 4.25-4.31 (m, 2H); 4.14 (dd, *J* = 8.2 and 6.0 Hz, 2H); 3.67 (dd, *J* = 8.2 and 6.5 Hz, 2H); 3.45 (dd, *J* = 11.7 and 5.8 Hz, 2H); 3.36 (dd, *J* = 11.7 and

7.1 Hz, 2H); 1.44 (s, 6H); 1.35 (s, 6H). ^{13}C NMR (100 MHz, CDCl_3); δ (ppm): 109.7, 77.6, 69.9, 27.1, 25.7, 8.7. MS: m/z (rel. int.) 488 ($\text{M}^+ - 2$, 1.0), 115 (32.1), 57 (64.8), 43 (100.0). HRMS (ESI): m/z calcd for $\text{C}_{12}\text{H}_{22}\text{O}_4\text{Te}_2$ [$\text{M}+\text{Na}$] $^+$: 508.9511; found: 508.9533.

(S,S)-bis(2,2-Dimethyl-1,3-dioxolanylmethyl)diselenide (4a): Yield: 0.124 g (75%, 99.9 ee); orangish oil; $[\alpha]_{\text{D}}^{20}$: - 39.11 (c 0.53, CH_2Cl_2). The characterization data from NMR, MS and HRMS spectra were identical in all aspects with those of (*R,R*)-(+)-**4a** enantiomer.

(S,S)-bis(2,2-Dimethyl-1,3-dioxolanylmethyl)disulfide (4b): Yield: 0.037 g (30%, 99.9 ee); colorless oil; $[\alpha]_{\text{D}}^{20}$: - 24.69 (c 0.27, CH_2Cl_2). The characterization data from NMR, MS and HRMS spectra were identical in all aspects with those of (*R,R*)-(+)-**4b** enantiomer.

(S,S)-bis(2,2-Dimethyl-1,3-dioxolanylmethyl)diteelluride (4c): Yield: 0.102 g (49%, 99.9 ee); reddish oil; $[\alpha]_{\text{D}}^{20}$: - 33.74 (c 0.39, CH_2Cl_2). The characterization data from NMR, MS and HRMS spectra were identical in all aspects with those of (*R,R*)-(+)-**4c** enantiomer.

2.2.3 Synthesis of (*R*)-3-(2,3-dihydroxypropylselenanyl)propanenitrile 6b: (i) Michael addition step: To a mixture of diselenide (*R*)-**4a** (0.390g; 1.0 mmol) in PEG-400 (4.0 mL) under N_2 atmosphere, NaBH_4 (0.046g; 1.2 mmol) was added at room temperature and the mixture stirred during 30 min. After, it was added acrylonitrile (0.053g; 1.0 mmol) and the mixture was stirred for additional 2 h at 50 °C. Then, the solution was cooled to room temperature, diluted with ethyl acetate (10.0 mL) and washed with water (3x 10.0 mL). The organic phase was separated, dried over MgSO_4 , and the solvent was evaporated under reduced pressure. The product was isolated by column chromatography using hexanes/ethyl acetate (80:20) as the eluent. Compound (*R*)-**6a** was characterized and the spectral data are listed below:

(*R*)-3-(2,2-Dimethyl-1,3-dioxolanylmethylselenanyl)propanenitrile (6a): Yield: 0.219 g (88%, 99.9 ee); colorless oil; $[\alpha]_{\text{D}}^{20}$: + 17.10 (c 0.57, CH_2Cl_2). ^1H NMR (CDCl_3 , 400 MHz): δ 4.31-4.37 (m, 1H); 4.11 (dd, J = 8.3 and 6.1 Hz, 1H); 3.68 (dd, J = 8.3 and 6.6 Hz, 1H); 2.93-2.78 (m, 6H); 1.44 (s, 3H); 1.36 (d, J = 0.5 Hz, 3H). ^{13}C NMR (100 MHz, CDCl_3); δ (ppm): 118.7, 109.6, 76.0, 69.2, 27.2, 26.8, 25.4, 19.6, 18.2. MS: m/z (rel. int.) 249 (2.1), 101 (31.5), 43 (100.0). Anal. Calcd for $\text{C}_9\text{H}_{15}\text{NO}_2\text{Se}$: C 43.56, H 6.09, N 5.64. Found: C 43.42, H 5.98, N 5.76.

2.2.4 Synthesis of (*R*)-3-(2,3-Dihydroxypropylselenanyl)propanenitrile (6b): (ii) Deprotection step: To a mixture of (*R*)-**6a** (0.125g; 0.5 mmol) in water (2.0 mL), ZnBr_2 (0.111g; 0.5 mmol) was added and the resulting solution was stirred for 24 h under reflux. Then, the solution was cooled to room temperature, received in brine (15.0 mL) and the product was extracted with ethyl acetate (3x 10.0 mL). The organic phase was separated, dried over MgSO_4 , and the solvent was evaporated under reduced pressure. The product was isolated by column chromatography using hexane/ethyl acetate (50:50) as the eluent. Compound (*R*)-**6b** was fully characterized and the spectral data are listed below:

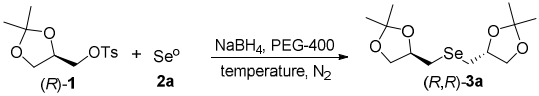
(*R*)-3-(2,3-Dihydroxypropylselenanyl)propanenitrile (6b): Yield: 0.041 g (40%, 99.9 ee); colorless oil; $[\alpha]_{\text{D}}^{20}$: - 22.64 (c 0.66, CH_2Cl_2). ^1H NMR ($\text{DMSO}-d_6$, 300 MHz): δ 3.98 (d, J = 5.0 Hz, 1H); 3.74 (t, J = 5.6 Hz, 1H); 2.67-2.77 (m, 1H); 2.37-2.48 (m, 2H); 1.98-2.02 (m, 2H); 1.84-1.91 (m, 3 H); 1.72 (dd, J = 12.5 and 6.8 Hz, 1H). ^{13}C NMR (75 MHz, $\text{DMSO}-d_6$); δ (ppm): 120.2, 71.6, 64.9, 28.3, 18.9, 18.0. MS: m/z (rel. int.) 209 (19.3), 133 (30.8), 54 (100.0). HRMS (ESI): m/z calcd for $\text{C}_6\text{H}_{11}\text{NO}_2\text{Se}$ [$\text{M}+\text{Na}$] $^+$: 231.9853; found: 231.9833.

3 Results and discussion

Our initial studies were focused on the development of an optimal set of reaction conditions to obtain compounds (*R,R*)- and (*S,S*)-**3a-c** starting from the respective (*R*)- or (*S*)-solketal tosylate **1**. To establish the best conditions, the (*R*)-solketal tosylate **1a** was reacted with elemental selenium **2a** in the presence of NaBH_4 as the reducing agent under N_2 atmosphere and using PEG-400 as the solvent (Table 1).

The influence of the temperature and the stoichiometric amounts of (*R*)-**1a**, **2a** and NaBH_4 were evaluated. It was found that using an equimolar amount of NaBH_4 and PEG-400 at room temperature, the reaction proceeds slowly, giving (*R,R*)-**3a** in 62% yield after 24 h (Table 1, entry 1). By increasing the temperature to 50 °C, the desired product **3a** was obtained in 84% yield after 3 h (Table 1, entry 2). The same result was obtained by rising the temperature to 80 °C (Table 1, entry 3). In another experiment, an excess of (*R*)-**1a** caused a decrease to 60% in the yield of (*R,R*)-**3a**, while when 0.6 mmol of **2a** was used, product was obtained in 82% yield (Table 1, entries 4-5). When ethanol or a mixture ethanol/THF 1:1 were used as solvent, low yields of (*R,R*)-**3a** were obtained (Table 1, entries 6-7).

Table 1. Optimization of the synthesis of (*R,R*)-bis(2,2-dimethyl-1,3-dioxolanylmethyl)selenide **3a**.^a

						
Entry	1a (mmol)	2a (mmol)	NaBH_4 (mmol)	Temperature (°C)	Time (h)	Yield (%) ^b
1	1.0	0.5	1.1	r.t.	24	62
2	1.0	0.5	1.1	50	3	84
3	1.0	0.5	1.1	80	3	83
4	1.2	0.5	1.1	50	3	60
5	1.0	0.6	1.3	50	3	82
6 ^c	1.0	0.5	1.1	50	5	46
7 ^d	1.0	0.5	1.1	50	6	62

^a A mixture of (*R*)-**1**, **2a** and NaBH_4 in PEG-400 (4.0 mL) under N_2 atmosphere was stirred for the indicated time. ^b Yields are given for isolated product (*R,R*)-**3a**. ^c Ethanol (4.0 mL) was the solvent. ^d A mixture of ethanol/THF 1:1 (4.0 mL) was used as the solvent.

ARTICLE

Journal Name

Table 2. Synthesis of enantiomerically pure chalcogenides **3a-c** and dichalcogenides **4a-c**.

Entry	Tosyl solketal 1	Y 2	Products 3 and 4	Time (h)	Yield (%) ^a
1		Se 2a		3.0	84
2	(<i>R</i>)- 1	S 2b		5.0	75
3	(<i>R</i>)- 1	Te 2c		3.0	60
4		2a		3.0	70
5	(<i>S</i>)- 1	2b		5.0	65
6	(<i>S</i>)- 1	2c		3.0	55
7	(<i>R</i>)- 1	2a		3.5	80
8	(<i>R</i>)- 1	2b		5.0	42
9	(<i>R</i>)- 1	2c		3.5	47
10	(<i>S</i>)- 1	2a		3.5	75
11	(<i>S</i>)- 1	2b		5.0	30
12	(<i>S</i>)- 1	2c		3.5	49

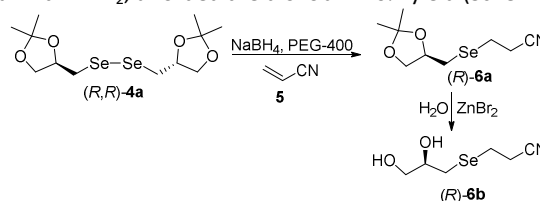
^a Yields are given for isolated products.

Once the best conditions for the synthesis of bis-(*R,R*)-(2,2-dimethyl-1,3-dioxolanylmethyl)selenide **3a** were established, the protocol was extended to the (*S*)-enantiomer of **1** and to other chalcogens (sulfur and tellurium). As indicated in Table 2, the respective chalcogenides were obtained in moderate to good yields (Table 2, entries 1-6). A small decrease in yields was observed when sulfur and tellurium were used in place of selenium. The enantiomeric bis-(*R,R*)- and bis-(*S,S*)-(2,2-dimethyl-1,3-dioxolanylmethyl)sulfides **3b** were obtained in 75 and 65% yields (Table 2, entries 2 and 5), while the telluride analogs (*R,R*)- and (*S,S*)-**3c** were isolated in 60 and 55% yields, respectively (entries 3 and 6). In the reaction using tellurium, a large amount of unreacted tellurium powder was recovered. In the case of the (*R,R*)- and (*S,S*)-sulfides **3b**, the formation of the respective disulfides (*R,R*)- and (*S,S*)-**4b** was observed in 10

and 8% yields. This concurrent reaction may explain the moderated yields of **3b**.

Because we observed the formation of diselenide during the preparation of **3b**, we decided to investigate the preparation of chiral dichalcogenides **4** selectively from (*R*)-**1** and elemental chalcogen. We started with the reaction of elemental selenium **2a** with NaBH₄ in EtOH to generate *in situ* Na₂Se₂,¹⁴ which was reacted with 2 equiv. of tosylate (*R*)-**1**. However, the results were not satisfactory, with the desired diselenide (*R,R*)-**4a** being obtained in only 12% yield after 3 h, instead forming predominantly selenide (*R,R*)-**3a**. Fortunately, (*R,R*)-bis(2,2-dimethyl-1,3-dioxolanylmethyl)diselenide **4a** was obtained in 80% yield when Li₂Se₂ was used (Table 2, entry 7). The Li₂Se₂ was generated according previously described by Lüdtkke and co-workers, who used super hydride (LiBHET₃) and elemental selenium to generate the nucleophilic selenium species.¹⁵ After stirring the mixture for 20 min under N₂, a solution of (*R*)-solketal tosylate **1** in THF was added and the stirring was continued for additional 3.5 h. Following, the same approach was applied using sulfur and tellurium, affording the desired dichalcogenides (*R,R*)-**4b** and (*R,R*)-**4c** in 42 and 47% yields, respectively (Table 2, entries 8-9). Similarly, starting from (*S*)-solketal tosylate **1**, the (*S,S*)-dichalcogenides **4a-c** were prepared in 75, 30 and 49% yields, respectively (Table 2, entries 10-12). The low yield observed for the disulfide **4b** was attributed to the formation of others side products, as trisulfide, formed by the reaction of Na₂S₃, generated *in situ* from the reaction of sulfur and LiBHET₃ (determined by GC/MS). In fact the generation of M₂S₂ (M = Na, K) from elemental sulfur is not trivial and is often accompanied by other polysulfides as co-products of the reduction.¹⁶ The low stability of Li₂Te₂ under our reaction conditions is responsible, at least in part, for the low yield of ditelluride **4c**, with a large amount of elemental tellurium being isolated at the end of the reaction.

We are interested in the synthetic applications of organic chalcogen compounds,¹⁷ as well as in the preparation of new selenium-containing bioactive molecules, e.g. antioxidants¹³ and antifungal agents.¹⁸ Thus, we explore the potential of the new chiral diselenide (*R,R*)-**4a** in the synthesis of the new, water-soluble and highly functionalized (*R*)-3-(2,3-dihydroxypropylselenanyl)propanenitrile **6b**. The first step of the reaction involved a Michael-addition of the selenolate anion, generated *in situ* from (*R,R*)-**4a**, to acrylonitrile **5**. The adduct (*R*)-**6a** was isolated in 88% yield and, after deprotection of the ketal with ZnBr₂, afforded the diol **6b** in 40% yield (Scheme 2).

**Scheme 2.** Synthesis of (*R*)-3-(2,3-dihydroxypropylselenanyl)propanenitrile **6b**.

4 Conclusion

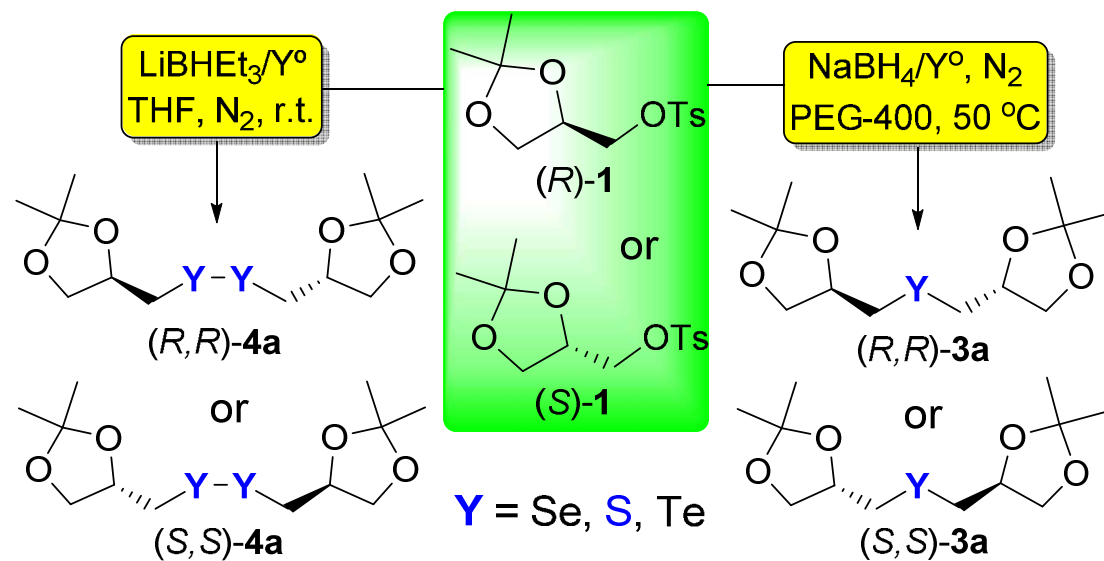
In conclusion, we have demonstrated the efficient synthesis of novel enantiomerically pure bis-(1,3-dioxolanymethyl)chalcogenides and dichalcogenides derivatives from glycerol. By this procedure, twelve new asymmetric chalcogenides (S, Se and Te) were prepared in reasonable to good yields after 3-5 h of reaction starting from chiral solketal tosylates and chalcogenolate anions. The selenium-derivatives **3a** and **4a** were obtained in higher yields in all the tested examples (70-84%), while (S,S)- and (R,R)-disulfides **4b** were isolated in 30 and 42% yields, respectively. The new water-soluble (R)-3-(2,3-dihydroxypropylselenanyl) propanenitrile **6b** was obtained in two steps from diselenide (R,R)-**4a** (35.2 % overall yield). The biological activities of this new class of compounds are now under investigation in our group.

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Notes and references

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Enantiomerically pure bis-1,3-dioxolanylmethyl chalcogenides and dichalcogenides (S, Se and Te) were prepared from chalcogenides and chiral solketal tosylates.